```
  GNU nano 7.2
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <time.h>

int main() {
    struct timespec start, end;
    pid_t jobs[3];

    clock_gettime(CLOCK_MONOTONIC, &start);

    for (int i = 0; i < 3; i++) {
        jobs[i] = fork();

        if (jobs[i] < 0) {
            perror("fork");
            return 1;
        }

        if (jobs[i] == 0) {
            execlp("sleep", "sleep", "2", NULL);
            perror("execlp");
            _exit(1);
        }
    }

    for (int i = 0; i < 3; i++)
        waitpid(jobs[i], NULL, 0);

    clock_gettime(CLOCK_MONOTONIC, &end);

    double elapsed = (end.tv_sec - start.tv_sec) +
        (end.tv_nsec - start.tv_nsec) / 1e9;

    printf("Three jobs completed.\n");
    printf("Execution time: %.2f seconds\n", elapsed);

    return 0;
}
```

```
 GNU nano 7.2
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    printf("Parent PID: %d\n", getpid());
    fflush(stdout);

    pid_t pid = fork();

    if (pid < 0) {
        perror("fork");
        return 1;
    }

    if (pid == 0) {
        printf("Child PID: %d\n", getpid());
        fflush(stdout);

        execlp("date", "date", NULL);

        perror("execlp");
        _exit(1);
    }

    int status;

    if (waitpid(pid, &status, 0) < 0) {
        perror("waitpid");
        return 1;
    }

    printf("Child process completed.\n");
    printf("Parent process completed.\n");

    return 0;
}
```

```
sravya@Sravya1955:~/os$ nano q23a.c
sravya@Sravya1955:~/os$ gcc -Wall -Wextra q23a.c -o q23a
sravya@Sravya1955:~/os$ ./q23a
Three jobs completed.
Execution time: 2.05 seconds
sravya@Sravya1955:~/os$ nano q23b.c
sravya@Sravya1955:~/os$ gcc -Wall -Wextra q23b.c -o q23b
sravya@Sravya1955:~/os$ ./q23b
Parent PID: 3040
Child PID: 3041
Sun Oct 11 06:06:40 UTC 2026
Child process completed.
Parent process completed.
sravya@Sravya1955:~/os$
```